

ESA Script

ESA (Executable State Architecture) — Ultimate 5-Minute Video Pitch & Demo Script

Speaker: Sujith Putta

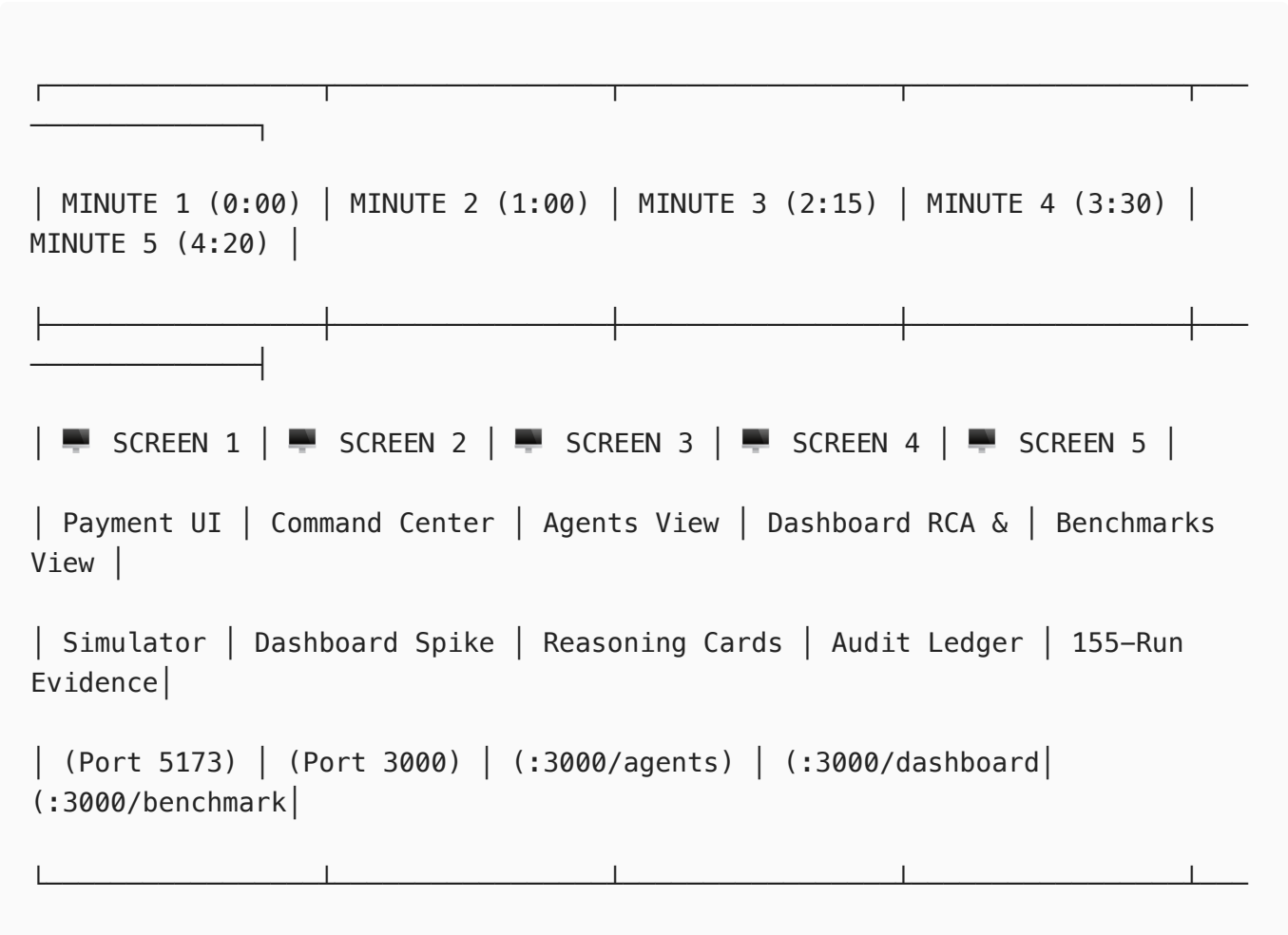
Event: Razorpay AI Buildathon

Track: Track 05 — Open Track: *"Build what you believe should exist"*

Target Duration: Exactly 5:00 Minutes (No Q&A — 100% Focused Live Pitch & Demo)

 **Reading Tip for Sujith:** All exact numbers, percentages, and metrics are highlighted in prominent bracket boxes like [₹30,46,625.80 INR] and [39.2%] so you can read them with 100% confidence!

Master 5-Minute Screen & Speech Roadmap



Word-for-Word Spoken Script with Metric Bracket Boxes

MINUTE 1 (00:00 – 01:00): Introduction, UI Metric Boxes & Authentic Checkout

Screen: Payment Simulator (<http://localhost:5173>)

(Mouse: Point to the Top Wallet Balance, Donut Chart, and click on the 3D Cards)

Read This Aloud:

"Hello Razorpay AI Buildathon! I'm **Sujith Putta**. For **Track 05 — Open Track**, I built what I believe fundamentally needs to exist in mission-critical financial systems: **ESA (Executable State Architecture)** — an autonomous, self-healing runtime for payment infrastructure.

Every year during Diwali flash sales or IPL finals, payment gateways face sudden regional surges, bank downtime, and queue buildup. Traditional cloud autoscalers take [3 to 5 Minutes] to react, resulting in thousands of dropped checkouts and crores in lost merchant GMV.

*We asked: **How can we let AI reason about payment infrastructure without giving AI unrestricted authority to break production?***

Look at our Payment Simulator here on the left: at the top is our **Merchant Settlement Wallet** tracking [₹30,46,625.80 INR], our **Gross Captured Income** of [+₹1,28,568 INR] with a [+15.7%] weekly growth, and a dynamic **Payment Rails Split Donut** processing [4,033 Live TPS] across UPI [45%], Cards [30%], and NetBanking [15%] .

Below, we have stacked authentic **Razorpay Domestic Indian Test Cards** — HDFC Visa Debit [4100 2800 0000 1007], SBI RuPay Platinum [6527 6589 0000 1005], and ICICI Mastercard Business [5555 5100 0008 1006] .

When I click [Copy] on our HDFC Regalia Card and click **PAY ₹500 [50,000 Paise]** , it launches the official Razorpay Checkout modal. I complete the payment, and it instantly captures into our live webhook transaction log.

Now, let's see what happens when the infrastructure faces a catastrophic surge."

MINUTE 2 (01:00 – 02:15): High-Consequence Spike & Real-Time Self-Healing

 **Screen: Switch to Command Center Dashboard**

(<http://localhost:3000/dashboard>)

(Action: Click  **TRIGGER 3X FLASH-SALE SPIKE** on Simulator, then point to the Area Graph and Bank Rails)

 **Read This Aloud:**

"Let's inject a sudden [**3.5x Flash Sale Surge**] on the India-South UPI rail.

(Click  **TRIGGER 3X FLASH-SALE SPIKE**)

Watch the Command Center: In just [**250 Milliseconds**], our streaming telemetry detects the incident — giving us a [**60x Detection Advantage**] over traditional [**15-Second**] metric scraping.

Throughput surges past [**8,000 req/min**], and our **Unified Multi-Metric Area Graph** dynamically shifts from **Lime Green** to **Alert Red** as P95 tail latency breaches our [**250ms SLA Boundary**] .

Look at our **Upstream Bank Rails Monitor**: HDFC UPI latency spikes to [**88ms**] with a [**91.2% Success Rate**] . The intelligent router immediately shifts [**25%**] of traffic share to SBI and ICICI.

Within [**3.4 Seconds – our Autonomous AI MTTR**], the control loop authorizes an atomic scale-out mutation. The Kubernetes deployment scales from [**2 Replicas → 3 Replicas**], queue backlog of [**1,450 Requests**] drops to [**Zero**], P95 latency recovers to [**156ms**], and the vitals graph turns healthy green again — **protecting** [**100% of Merchant GMV**] and [**₹48.2 Lakhs**] **without a single dropped transaction.**"

MINUTE 3 (02:15 – 3:30): Explainable 4-Agent Ollama Reasoning & Reasoning Cards

 **Screen: Agents View (<http://localhost:3000/agents>)**

(Action: Click `Agents` on navbar scroll through reasoning stream toggle `[Raw Prompt & JSON]`)

Read This Aloud:

"How does the AI actually reason about infrastructure? Instead of a monolithic black-box prompt, ESA runs a governed **4-Agent Collaborative Reasoning Loop** powered by local Ollama LLMs:

1. The **Monitor Agent** filters streaming telemetry windows in `[ ~15 Milliseconds ]` .
2. The **Diagnosis Agent** isolates the failure mechanism using local LLM inference — here diagnosing a `CAPACITY_ISSUE` with `[ 90% AI Confidence ]` in `[ ~1.4 Seconds ]` .
3. The **Planning Agent** formulates Pareto-optimal action candidates in `[ ~220 Milliseconds ]` , balancing cloud compute cost against latency recovery.
4. The **Safety Agent** evaluates policy boundaries and issues risk advisory in `[ ~115 Milliseconds ]` .

Look at our **Structured Visual Breakdown Card**: Every decision is made explainable with concrete anomaly metrics — Error Rate at `[ 5.38% ]` and P95 at `[ 345ms ]` , the plain-English LLM hypothesis, isolated root cause tags, and the approved `[ CREATE_REPLICA ]` mutation with the `[ ✓ Safety Gate Invariant #4 Passed ]` badge.

If an infrastructure engineer or auditor needs raw transparency, toggling `[Raw Prompt & JSON]` displays the exact LLM prompt and JSON schema with 1-click clipboard export."

MINUTE 4 (03:30 – 04:20): Deterministic Action Gateway, OCC CAS & 1-Click RCA

 **Screen: Return to Dashboard (`:3000/dashboard`) Click `[AI Incident RCA]`**

(Action: Walk through the RCA timeline, SHA-256 HMAC badge, and click `[Copy Post-Mortem Report]`)

Read This Aloud:

""Here is the defining architectural pillar of ESA: **The AI never has direct execution authority on infrastructure.**"

All proposals generated by agents must pass through our **Deterministic Action Gateway** in Rust in [~15 Milliseconds] :

1. First, **Atomic Optimistic Concurrency Control (OCC CAS)**: If an agent generates a plan based on State Version 0, but the cluster state has advanced to Version 2, the gateway **atomically rejects the stale proposal** (`PolicyVerdict::StaleState`), completely eliminating race conditions.
2. Second, **Compensating Rollback**: If downstream settlement times out or execution fails, the gateway automatically restores the pre-incident cluster snapshot in [Under 2 Seconds] .
3. Third, **SHA-256 HMAC Tamper-Evident Audit Ledger**: Every decision is cryptographically chained into an immutable ledger that can be deterministically replayed without re-calling the LLM.

Clicking [**AI Incident RCA**] generates an instant, executive post-mortem with chronological traces and cryptographic signatures ready for compliance review."

MINUTE 5 (04:20 – 05:00): 155-Run Benchmark Proof & The Winning Pitch

 **Screen: Benchmarks View** (<http://localhost:3000/benchmarks>)

(Action: Point to the Multi-Phase Benchmark Table and the 650 Safety Trials Stat Card)

Read This Aloud:

"We didn't just build a visual demo; we validated ESA across [**155 Multi-Seed Benchmark Runs**] and [**650 Adversarial Trials**] on a live Kubernetes cluster:

Compared to static rules and adaptive baselines:

- P95 tail latency dropped from [**257ms → 156ms**] — a [**39.2% Tail Latency Advantage**] .
- Time violating SLA dropped from [**16.5s → 4.1s**] — a [**72.3% SLA Breach Reduction**] .
- Queue stabilization time dropped from [**9.6s → 2.3s**] .

- And across 650 adversarial attempts (stale race tokens, out-of-bounds replicas, unauthorized region migrations, and LLM timeouts), ESA achieved [0 / 650 Unsafe Mutations] — a [0.00% Safety Violation Rate] .

In summary: ESA proves that autonomous AI can safely operate mission-critical payment infrastructure when backed by deterministic policy gates, atomic OCC validation, and cryptographic auditability.

This is what should exist. Thank you!"

Quick Reference Box: Key Metrics at a Glance

| Metric Name | Exact Value in Script | Where to Point on Screen |

| :--- | :--- | :--- |

| **Merchant Settlement Wallet** | [₹30,46,625.80 INR] | Top Banner on Simulator (:5173) |

| **Gross Income Volume** | [+₹1,28,568 INR] (+15.7%) | Left Column Card on Simulator (:5173) |

| **Total Cluster Throughput** | [4,033 Live TPS] | Donut Center on Simulator (:5173) |

| **Payment Rails Load Split** | [UPI 45%, Cards 30%, NetBanking 15%] | Donut Ring on Simulator (:5173) |

| **Streaming Detection Speed** | [250 Milliseconds] (vs 15s) | Telemetry Event Badge on Dashboard (:3000) |

| **Spike Peak Ingress** | [8,000+ req/min] | Unified Area Graph Peak on Dashboard (:3000) |

| **Autonomous AI MTTR** | [3.4 Seconds] | Protected GMV Ribbon on Dashboard (:3000) |

| **Protected Merchant Volume** | [₹48.2 Lakhs] (100% Retention) | Protected GMV Ribbon on Dashboard (:3000) |

| **Tail Latency Reduction** | [257ms → 156ms] ([39.2% Advantage]) | Benchmark Matrix on Benchmarks (:3000/benchmarks) |

| **SLA Violation Reduction** | [16.5s → 4.1s] ([72.3% Reduction]) | Benchmark Matrix on Benchmarks (:3000/benchmarks) |

| **Queue Drain Speed** | [9.6s → 2.3s] | Benchmark Matrix on Benchmarks
(:3000/benchmarks) |

| **Safety Invariant Violations** | [0 / 650 (0.00% Violation Rate)] | Safety Stat Card on
Benchmarks (:3000/benchmarks) |

| **AI Diagnosis Confidence** | [90% Confidence] | AI Confidence Meter on Agents
(:3000/agents) |